

Memo

Note Title

17 May 2024

Section A

$$1) t'(x) = 2e^{2x}, \quad t'(\ln 3) = 2e^{2\ln 3} = 2e^{\ln 9} = 18$$

$$2) m''(x) = \frac{2}{1+x^2} - \frac{2x^2}{(1+x^2)^2}$$

$$m''(1) = \underline{\underline{\frac{1}{2}}}$$

$$3) \frac{dy}{dx} = \frac{1}{2\sqrt{x}} - 1, \quad \left. \frac{dy}{dx} \right|_{x=1} = -\frac{1}{2}. \quad \text{Now } y = -\frac{1}{2}x + C$$

$\begin{aligned} 0 &= -\frac{1}{2}(1) + C \\ &= -\frac{1}{2} + C \\ \text{and } C &= \underline{\underline{\frac{1}{2}}} \end{aligned}$

$$4) (1) f'(x) = \frac{d}{dx}(\ln 3 + \ln x^3) = \frac{d}{dx}(\ln 3 + 3\ln x) = 0 + \frac{3}{x}$$

$$(2) f(x) = \sinh(3x), \quad f'(x) = 3\cosh 3x, \quad f''(x) = 9\sinh 3x$$

$$(3) f(x) = e^{3x}, f'(x) = 3e^{3x}$$

$$f'(-x) = 3e^{-3x}$$

Statements 1 and 3 are correct.

$$5) \lim_{x \rightarrow \infty} (\sqrt{x^2 + 2x + 1} - x) = \lim_{x \rightarrow \infty} \sqrt{(x+1)^2} - x$$

$$= \lim_{x \rightarrow \infty} x+1 - x$$

$$= 1$$

Since $x \rightarrow \infty$,
 $\sqrt{(x+1)^2} = x+1.$

6) Statements 3 and 4 are correct.

7) Only statement 3 is correct.

$$8) g'(1) = \lim_{x \rightarrow 1} \frac{g(x) - g(1)}{x - 1}. \quad g'(x) = \frac{1}{1+x^2} \Rightarrow g'(1) = \frac{1}{2}.$$

Section B

$$9.1) f'(x) = \frac{2x}{\sqrt{1-x^4}}$$

$$9.2) f'(x) = \frac{\frac{1}{2} \cdot x - \ln x}{x^2}$$

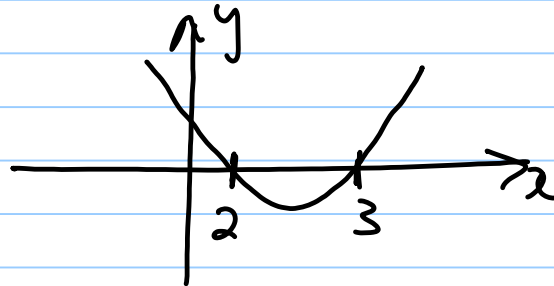
$$9.3) \sec^2(\tan(\tan x)) \sec^2(\tan x) \sec^2 x.$$

$$10) 1 - \frac{1}{x-2} < 0$$

$$\Rightarrow \frac{x-2-1}{x-2} < 0$$

$$\Rightarrow \frac{x-3}{x-2} < 0$$

$$\Rightarrow (x-3)(x-2) < 0$$



$$\Rightarrow x \in (2, 3)$$

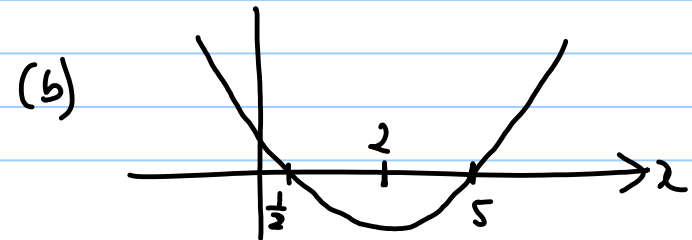
$$11) f'(x) = \cos x - \sin x = 0$$

$$\Rightarrow \tan x = 1$$

$$\Rightarrow x = \frac{\pi}{4} \text{ or } x = \frac{5\pi}{4}.$$

Critical nr's are $\frac{\pi}{4}$ and $\frac{5\pi}{4}$.

12) See textbook p. 291.



Section C

14) If $y = x^{\sin x}$

Consider $\ln y = \ln x^{\sin x} = \sin x \ln x$

$$\frac{d}{dx}(\ln y) = \frac{d}{dx}(\sin x \ln x)$$

$$\frac{1}{y} \frac{dy}{dx} = \cos x \ln x + \frac{\sin x}{x}$$

$$\frac{dy}{dx} = x^{\sin x} \left(\cos x \ln x + \frac{\sin x}{x} \right)$$

$$15) \lim_{x \rightarrow 1^+} \left(\frac{2}{x^2-1} - \frac{1}{x-1} \right) = \lim_{x \rightarrow 1^+} \frac{2 - (x+1)}{(x-1)(x+1)} = \lim_{x \rightarrow 1^+} \frac{1-x}{(1+x)(x-1)} = \lim_{x \rightarrow 1^+} \frac{-1}{x+1} = \frac{-1}{2}$$

Question 16.

$$\lim_{x \rightarrow -\infty} h(x) = \lim_{x \rightarrow -\infty} \frac{x}{\sqrt{3x^2 + 1}} \left(\frac{-\infty}{\infty} \right) = \frac{-1}{\sqrt{3}}$$

See example 4 in textbook,
p. 131.

$y = \frac{-1}{\sqrt{3}}$ is a horizontal asymptote.

$$\lim_{x \rightarrow \infty} \frac{x}{x+7} \left(\frac{\infty}{\infty} \right)$$

$$= \lim_{x \rightarrow \infty} \frac{1}{1 + 7/x}$$

$$= 1$$

$\Rightarrow y = 1$ is a horizontal asymptote.

(conclusions must be given.)

Question 17

$$-1 \leq \sin \frac{1}{x^2} \leq 1 \text{ for all } x \in \mathbb{R} \setminus \{0\}$$

$$\Rightarrow -x^2 \leq x^2 \sin \frac{1}{x^2} \leq x^2$$

$$\lim_{x \rightarrow 0} -x^2 = 0$$

$$\lim_{x \rightarrow 0} x^2 = 0$$

} by the squeeze theorem: $\lim_{x \rightarrow 0} x^2 \sin \frac{1}{x^2} = 0 = \underline{\underline{f(0)}}$

$\Rightarrow \underline{\underline{f \text{ is continuous at } x=0.}}$

MS

Question 18

$$\frac{d}{dx}(xy-2) = \frac{d}{dx} \ln(x^2+y-4)$$

$$\Rightarrow 1 \cdot y + x \frac{dy}{dx} = \frac{2x + \frac{dy}{dx}}{x^2+y-4}$$

Now substitute y with $f(x)$ and $\frac{dy}{dx}$ with $f'(x)$.

Use the fact that $f(2) = 1$

and find $f'(2) = 3$.

Question 19

See textbook p. 157-158.

Question 20

See example 5 on p. 296.

You need to indicate that you use the Mean Value Theorem, as well as why you can use it.